

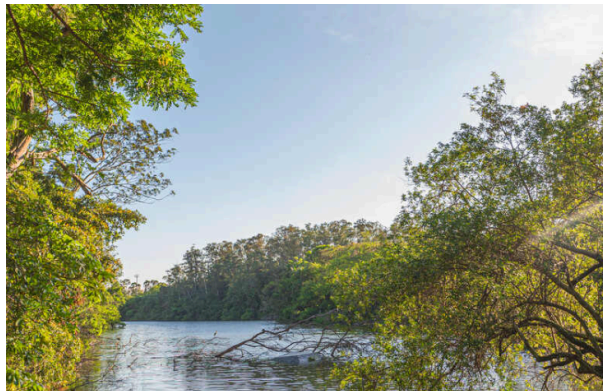
Chapter 19

Chapter Concepts

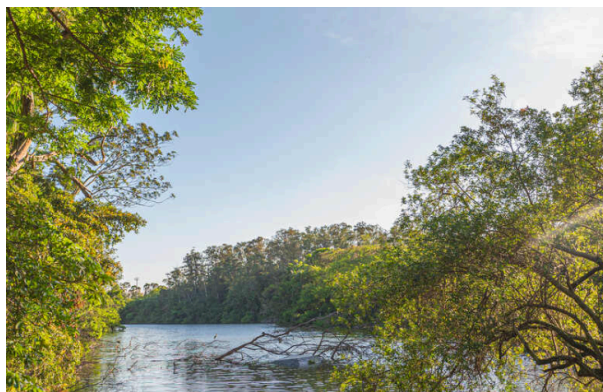
- **Sere**: All the stages of **successional change**.
- **Succession**: Community change following either **habitat disturbance** or the formation of new habitat.
- **Climax community**: The ultimate association of plants and animals in a sere.
- **Facilitation**, **inhibition**, and **tolerance** are key processes in succession.
- Succession becomes **self-limiting** as it approaches the **climax**.

Introduction and Krakatau Case Study

- **Ecological succession** is the process by which communities recover after disturbance, as seen in the Krakatau eruption, where initial colonization was dominated by sea- and wind-dispersed plants, followed by animal-dispersed species as succession progressed.
- The changing composition of plant colonists on Krakatau is visualized in the graph below, showing the shift from physical to animal dispersal over time.



- Succession involves a predictable sequence of species replacement, leading to a mature, stable community. The long-term development of an oak–hornbeam forest illustrates this gradual process.

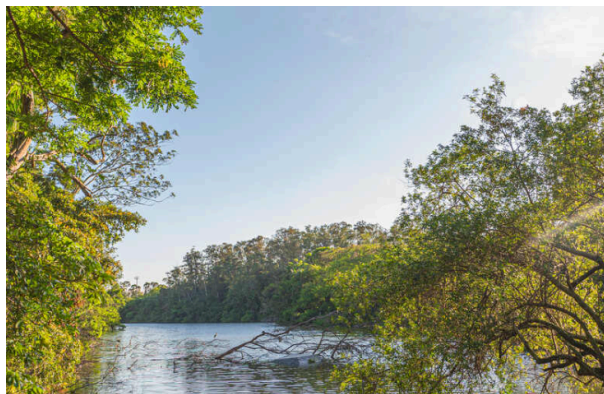


Stages and Types of Succession

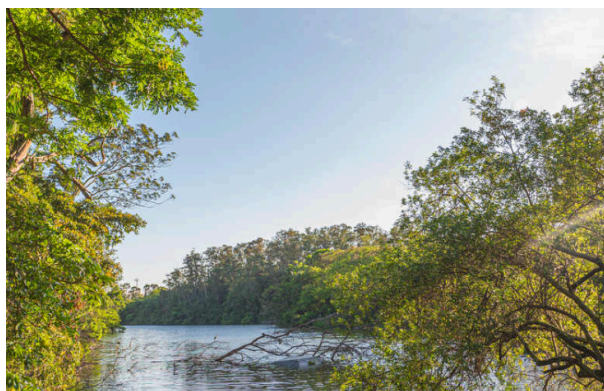
- In abandoned agricultural fields, **secondary succession** begins with annual plants, which are replaced by perennials, shrubs, and eventually trees, demonstrating how disturbance resets community structure.



- **Primary succession** occurs on substrates that have never supported life, such as sand dunes. Perennial grasses stabilize the sand, enabling shrubs and trees to establish in later stages.



- Wetland succession involves aquatic plants and sedges filling open water, leading to peat accumulation and transformation into terrestrial habitat.

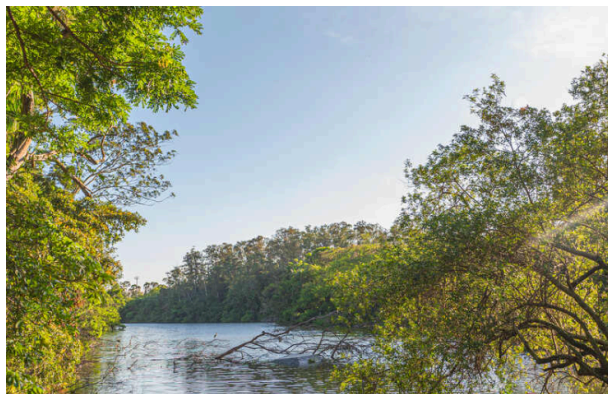


- The buildup of organic matter in peat bogs is a result of long-term succession, as shown by the accumulation of plant detritus in soil layers.

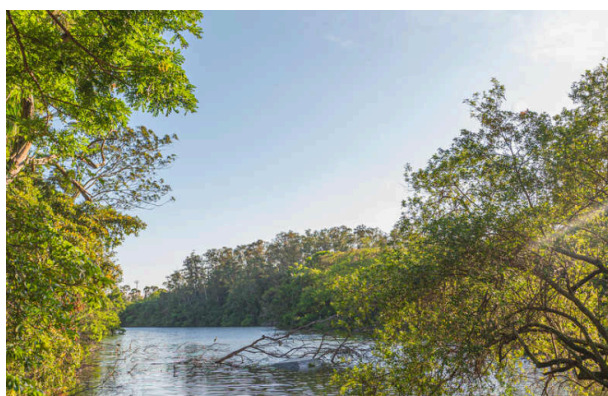


Disturbance, Gap Size, and Community Dynamics

- In marine and terrestrial systems, the size and isolation of disturbance gaps influence which species can colonize and persist. Limpets grazing on mussel beds create open patches, shaping community structure through biotic interactions.



- The life history traits of colonists, such as colonizing ability and competitive strength, determine successional outcomes. The table below summarizes these differences for sessile marine invertebrates.



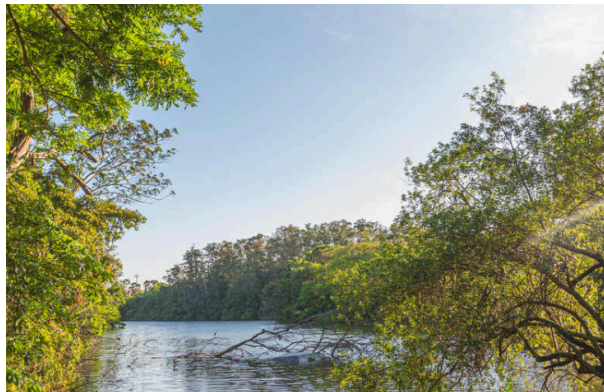
Climax Community and Environmental Gradients

- The **climax community** is not always a fixed endpoint; instead, communities may form a continuum along environmental gradients like soil moisture and pH. The composition of tree species in Wisconsin forests varies continuously, as shown in the continuum index.



Processes Governing Succession

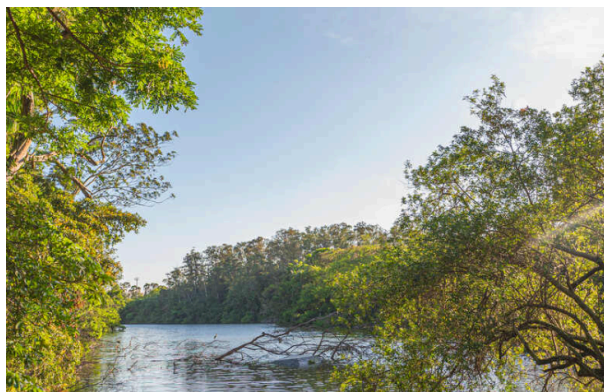
- **Facilitation** occurs when early colonists modify the environment to benefit later arrivals, such as alder trees enriching soils with nitrogen and enabling spruce establishment.



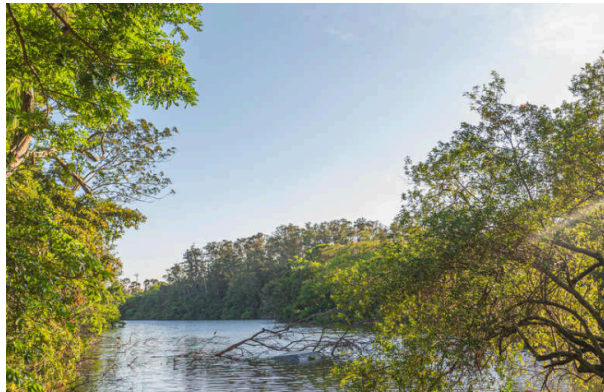
- **Inhibition** and **priority effects** can prevent new species from establishing, while **tolerance** allows some species to establish regardless of others, depending on their ability to survive prevailing conditions.

Life History Strategies and Succession

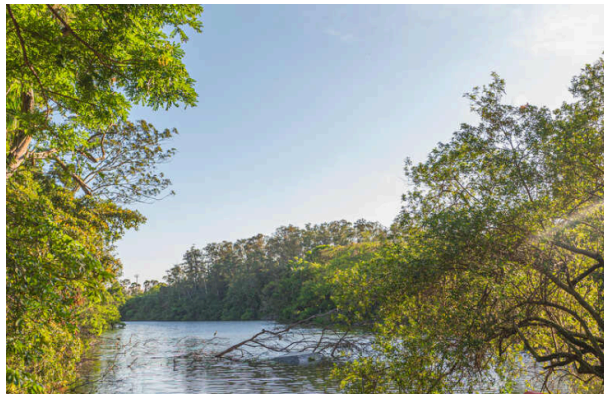
- Early successional species produce many small, wind-dispersed seeds and grow rapidly in open conditions, while late successional species have fewer, larger seeds and greater shade tolerance. The timing of germination and flowering for early successional plants is shown below.



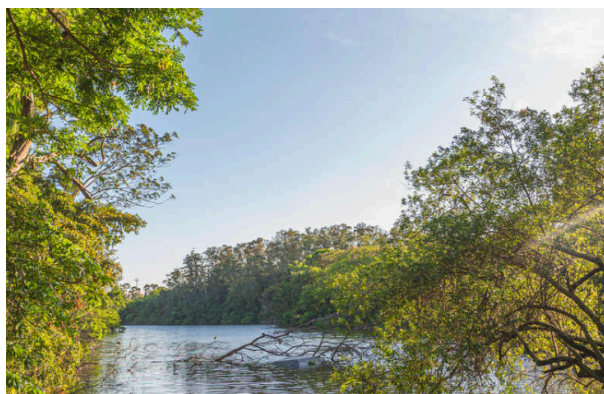
- Interactions with soil organisms, such as mycorrhizal fungi, influence which species succeed. The effects of these fungi on plant growth are highly variable and depend on local adaptation.



- Early and late successional plants differ in seed size, dispersal, growth rate, and shade tolerance, affecting their roles in succession. The table below compares these traits.



- Large-seeded species are more likely to survive in shaded environments, as shown by the relationship between seed weight and seedling mortality.



Self-Limiting Nature of Succession and Climax Replacement

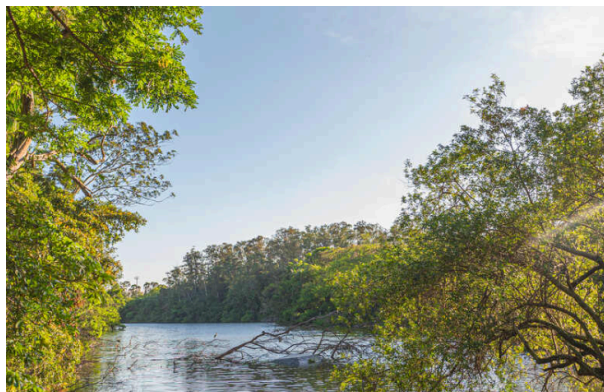
- As succession progresses, the addition of new species slows and the community approaches a steady state. Even mature forests continue to change as species like beech

and maple gradually replace oaks and hickories, reflecting ongoing competition and adaptation.



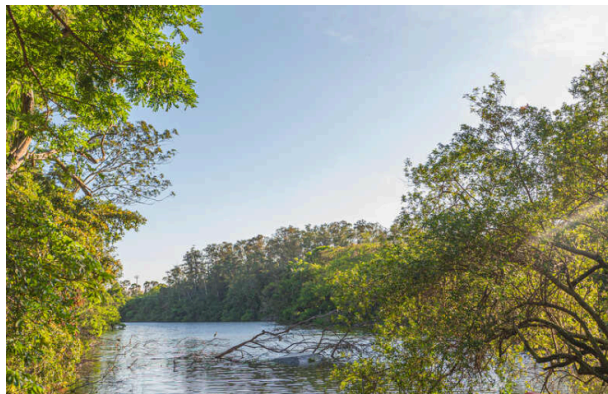
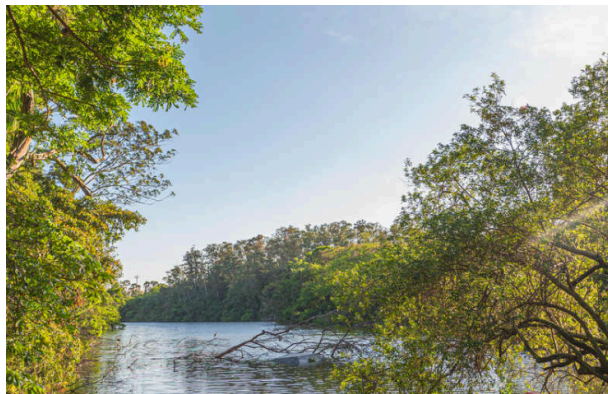
Fire, Grazing, and Alternative Stable States

- Fire-adapted species, such as longleaf pine, persist in environments with frequent burns, while grazing by large herbivores shapes the composition of grasslands and savannas.



Transient and Cyclic Climax Communities

- Some climax communities are transient, such as those in seasonal ponds or on animal carcasses, and must be re-established repeatedly. Others, like fir forests and heathlands, experience cyclic succession driven by environmental stress, resulting in shifting mosaics of vegetation.



Summary of Key Terms

- Succession, sere, climax community, facilitation, inhibition, tolerance, primary succession, secondary succession, seed bank, continuum, alternative climax, transient climax, cyclic succession.